MACHINE LEARNING IN PHYSICS

PROJECTS

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Project Description

The project has 2 parts:

1. Use a machine-learning model of your choice to solve a problem of interest to you. This could be, but does not have to be, one of the following classes of problem: classification, regression, data compression, structure detection, or solving an ordinary differential equation.

2. Document the project in a jupyter notebook.

Project Description

Problem Choose a problem from your field (if you're a graduate student) or you may tackle one of the following problems:

1. Jet classification (gluon vs. quark-initiated jets)

2. Nuclear properties modeling (AME 2020).

3. Solve the Friedmann equation for the $\Lambda\mathrm{CDM}$ model.

Jet Classification

Problem 1: Using 2D images of jets of particles, classify them into two classes: jets initiated by gluons or by quarks.

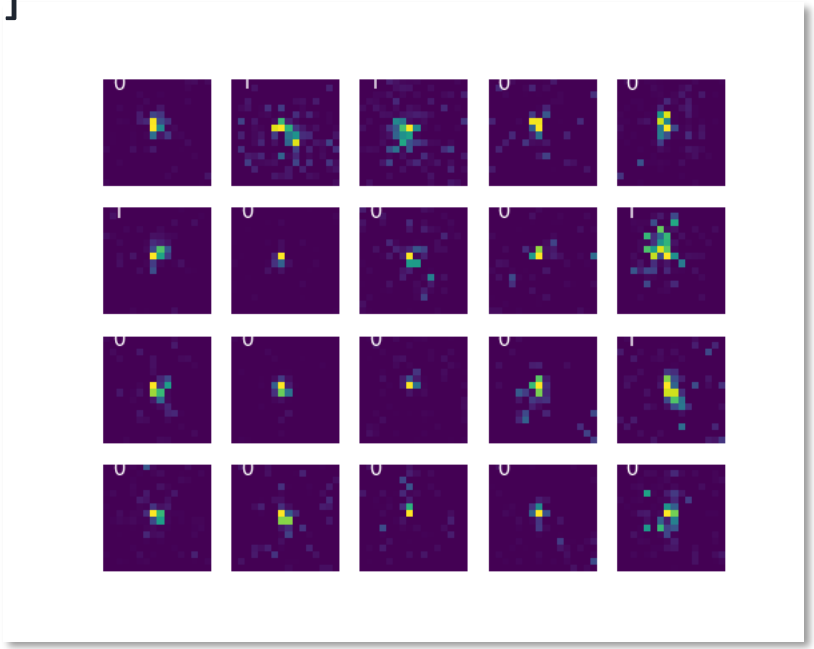
Dataset: Images of shape (1, 16, 16), which have been divided into: ['test_x', 'test_y', 'train_x', 'train_y', 'valid_x', 'valid_y']

Filename: jets.h5

len(train_x): 50,000

len(valid_x): 5000

len(test_x): 5000



Nuclear Properties

Problem 2: Predict a nuclear property given N, Z, and A.

Dataset: 3,558 nuclei from the AME 2020 mass data[1].

	name	N	Z	A	Mexcess	eMexcess	bindingE	ebindingE	betaE	ebetaE	atomicM	eatomticM
0	n	1	0	1	8.071318	4.4e-07	0.0	0.0	0.782347	4e-07	1.008665	4.7e-07
1	H	0	1	1	7.288971	1.3e-08	0.0	0.0	-999	-999	1.007825	1.4e-08
2	H	1	1	2	13.135723	1.5e-08	1.112283	2e-07	-999	-999	2.014102	1.5e-08
3	H	2	1	3	14.949811	8e-08	2.827265	3e-07	0.018592	6e-08	3.016049	8e-08
4	He	1	2	3	14.931219	6e-08	2.57268	1.5e-07	-999	-999	3.016029	6e-08
5	Li	0	3	3	-999	-999	-999	-999	-999	-999	-999	-999
6	H	3	1	4	24.621129	0.1	1.720449	0.025	22.196213	0.1	4.026432	0.107354
7	He	2	2	4	2.424916	1.5e-07	7.073916	2e-07	-22.9	0.212132	4.002603	1.6e-07
8	Li	1	3	4	25.32319	0.212132	1.15376	0.053033	-999	-999	4.027186	0.227733
9	H	4	1	5	32.892447	0.089443	1.336359	0.017889	21.661213	0.091651	5.035311	0.09602

Filename: AME2020.csv

1. The AME 2020 atomic mass evaluation, Chinese Phys. C **45**, 030002 (2021) and Chinese Phys. C **45**, 030003 (2021).

Nuclear Properties

Problem 2: Predict a nuclear property given N, Z, and A.

Dataset: 3,558 nuclei from the AME 2020 mass data[1].

Energy Variables:

Mexcess, bindingE, betaE: in MeV

atomicM: relative atomic mass

Note: entries in the CSV file with -999 implies missing data.

Friedmann Equation

Problem 3: Solve the Friedmann equation using a PINN.

Data set: To be generated by you!

Equation:

$$\frac{da}{dx} = a\sqrt{\Omega(a)}, \qquad x \equiv H_0 t$$

where for the ΛCDM model,

$$\Omega(a) = \frac{\Omega_M}{a^3} + \frac{(1 - \Omega_M - \Omega_\Lambda)}{a^2} + \Omega_\Lambda$$

Initial condition: $a(x = 0) = 0$.

Suggested domain: $(x, \Omega_M, \Omega_\Lambda) = (0, 1.5) \otimes (0, 1) \otimes (0, 1)$ with the constraint $\Omega(a) > 0$.

Writeup in Jupyter Notebook

Your write-up, using markdown, in the jupyter notebook must include the following elements and sections:

1. A Title
2. An Abstract
3. Introduction
4. Dataset Description
5. Model Description
6. Experiments
7. Summary

Ground Rules

1. You are free to help each other, but your paper must be your own.

2. You are free to use any resource, but you must cite all of them. If you use AI, you must explain in detail how.

3. You must document the data used, the architecture of your model, the training protocol and the results.

4. I should be able to run your code with standard Python modules.

Deadline

Last day of classes @ 11:59PM